GRU-Based Forecasting Outperforms LSTM and Bi-LSTM for Apple Stock and Bitcoin Prices

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***Abstract*—Accurate forecasting of highly volatile financial markets such as stocks and cryptocurrencies remains a significant challenge. This work compares three recurrent neural network architectures—Long Short-Term Memory (LSTM), Bidirectional LSTM (Bi-LSTM), and Gated Recurrent Unit (GRU)—in forecasting the prices of Apple Inc. (AAPL) stock and Bitcoin (BTC). Historical market data were collected through the yfinance API, and performance was measured using Root Mean Squared Error (RMSE), precision, recall, and F1-score. The results demonstrate that the GRU model achieved superior performance compared to both LSTM and Bi-LSTM, achieving the lowest RMSE (0.0152) and highest precision (0.99 for BTC, 0.98 for AAPL), while also offering superior computational efficiency with fewer parameters and faster inference times. These results highlight the GRU as a reliable and effective deep learning model for forecasting financial time series.**

***Index Terms*—GRU, LSTM, Bi-LSTM, financial time series forecasting, yfinance**

I. Introduction

Cryptocurrency, initially released with Bitcoin (BTC) in 2009, is a peerless form of virtual currency that functions independently of banks or government agencies and functions on the principle of blockchain technology for safe and clear transactions. Although its applicability is worldwide today, Bitcoin is very volatile in nature, with price rates often fluctuating enormously in a span of minutes, for instance, rendering forecasting difficult in most cases. Stock markets, including Apple Inc. (AAPL), too, presents intricate and dynamic price patterns determined by a series of economic and monetary factors. Conventional statistical methods often struggle to capture the nonlinear and noisy nature of financial data. This trend has encouraged wider use of deep learning approaches in forecasting financial time series. Among recurrent neural networks, Long Short-Term Memory (LSTM), Bidirectional LSTM (Bi-LSTM), and Gated Recurrent Units (GRU) are widely used due to their ability to learn temporal dependencies effectively. Although LSTM and Bi-LSTM are comprehensively studied, GRU offers a compact structure with fewer parameters and a higher training velocity with similar or superior accuracy. Following this approach, the current study evaluates and compares GRU, LSTM, and Bi-LSTM models for forecasting the prices of Apple stock and Bitcoin. Model performance is assessed using Root Mean Squared Error (RMSE) along with precision, recall, and F1-score. The results indicate that the GRU model achieves the highest overall predictive accuracy, demonstrating its effectiveness for financial forecasting applications. Instead than putting out a novel deep learning architecture, this work is positioned as a comparative analysis. The purpose is to analyze model behavior under consistent situations, not to bring methodological originality.

The project intends to solve three essential research questions: (i) Which of the recurrent neural network, or RNN, option—LSTM, Bi-LSTM, or GRU performs the best in estimating highly volatile investments such as bitcoin (Bitcoin) and Apple (AAPL) stock? (ii) How does complexity of model, characterized by layer depth and parameters count, effect the efficiency of computation during training and inference? (iii) How well can these models forecast directional movement

(classification-based trend prediction) as well as numerical prices (regression accuracy)? The key contributions of this study are as follows:

1) Using the same data, preprocessing, and training configurations, we give a thorough comparison of GRU, LSTM, and Bi-LSTM for BTC and AAPL predictions.
2) The proposed GRU-based forecasting model demonstrates superior computational efficiency by utilizing a simplified architecture with fewer gating mechanisms (update and reset gates) and trainable parameters compared to LSTM and Bi-LSTM variants.
3) We experimentally demonstrate efficiency differences throughout the three RNN variations instead of suggesting an innovative framework.

This paper is organized into five sections: Section II reviews related work, Section III describes the methodology, Section IV presents results and discussion, and Section V concludes with future directions.

II. Related Works

Deep learning networks were recently popularized in forecasting financial time series, including LSTM, GRU, and Bi-LSTM models. Corizzo and Rosen [1] embedded stock indices in a stacking set of textual characteristics and demonstrated high predictive power, although they relyed greatly on the quality of the text preprocessing. Gao et al. [2] have compared optimized LSTM and GRU models on the Chinese stock market and recorded GRU's slightly higher accuracy, emphasizing its performance on small data sets. Similarly, Siami-Namini et al. [3] suggested the potential of Bi-LSTM to outperform the usual LSTM on generic time series prediction tasks, although they did not carry out any finance-specific trial. Also proposed were models of hybrid and attention type. Lu et al. [4] introduced a hybrid model combining CNN, Bi-LSTM, and an attention mechanism on Shanghai Stock Exchange and attained higher-quality performance than single LSTM but with the payback of higher complexity in models. While Shaikh et al. [5] proposed a comparative study of RNN, LSTM, Bi-LSTM, GRU, and Transformer models for financial market forecasting, demonstrating that GRU and Transformer architectures achieve superior prediction accuracy across stock price datasets. Sudiatmika et al. [6] compared LSTM and GRU for gold price forecasting using Yahoo Finance data (2017–2024). Their results showed that GRU outperformed LSTM with lower error rates and higher accuracy. Viet et al. [7] presented an initial comparative analysis of LSTM and Bi-LSTM on Vietnamese stock data and decided in favor of Bi-LSTM in most cases, although their data set was quite narrow in focus. In addition, Gandhmal and Kumar [8] wrote a survey on forecasting approaches for stock markets, including a comprehensive comparison of approaches, and Al Qaisi et al. [9] conducted a traditional econometric analysis of stock price variables in the Amman Stock Exchange, providing a useful benchmark for comparing deep learning approaches. Specifically, they evaluated ARIMA alongside the GRU, LSTM, and Bi-LSTM models for financial time series prediction, Pirani et al. [10] revealed that Bi-LSTM predicted more accurately than the rest, while ARIMA was the least accurate. Generally, available works demonstrate that models' performance is highly dependent on the data set, feature, and measure. Although LSTM and Bi-LSTM are widely used, recent works suggest the potential for GRU to surpass them in certain works on financial prediction, motivating our experiment on Apple stock and Bitcoin price forecasting.

III. Methodology

The architecture of the proposed system is illustrated in Figure 2 and comprises four components: i) data pre-processing, ii) model training, iii) evaluation and iv) performance analysis. The preprocessing pipeline is illustrated in Figure 1

A. Data Preprocessing

Historical price data for BTC and AAPL were acquired using the yfinance API, which gives official daily OHLCV data. Only yfinance was utilized to maintain uniformity across both assets. When created, all historical data has been saved to finance. It records all the ups and downs of each and every crypto coin. So, we collected data from finance and organized all the data. Data are stored in time bases, like 1DAY, 7DAY, 1 MONTH, 1 YEAR. Using sliding windows, the dataset was converted into a supervised learning structure, where the past 30 days of data were used to predict the next day. WE organize all data for better useful in GRU. All data converted into a supervised learning format using sliding windows. The dataset was separated chronologically into 70% training, 15% validation, and 15% test sets to prevent temporal data leaking. To make sure that future data was never utilized during training, the sliding-window transformation (30 previous days $\rightarrow$ next-day prediction) was used after splitting. Early halting and hyperparameter adjustment were done using the validation set.

The training architecture is shown in Figure 2

B. Proposed model: GRU

A GRU-based neural network was designed with input, hidden, and output layers. The model was trained using the Mean Squared Error (MSE) loss function and optimized via the Adam optimizer. The Gated Recurrent Unit (GRU) is a type of recurrent neural network that is specifically designed in such a way that it works well when it is applied to sequences of data in order to extract temporal information. The GRU features a streamlined architecture compared to Long Short-Term Memory (LSTM) networks, incorporating only two gates: the update gate and the reset gate thus, it does not have a dedicated memory cell. It minimizes complexity in a manner such as a lesser number of trainable parameters, fast convergence, low memory requirements, and similar or better performance for most forecasting tasks. A GRU neural network was applied in this experiment with input, hidden, and output layers, trained on past Apple (AAPL) and Bitcoin (BTC) prices acquired through the yfinance API. It was optimized employing Adam

TABLE I: Summary of Related Works

Ref.	Year	Dataset	Method	Result	Limitations
[1]	2024	Stock indices + News headlines	Stacking ensemble (ML + NLP)	Improved accuracy when combining news with prices	High dependence on quality of text preprocessing
[2]	2021	Chinese stock market	Optimized LSTM, GRU	GRU slightly outperformed LSTM in MSE	Limited dataset, no external features
[3]	2019	Synthetic + real time-series	LSTM, Bi-LSTM	Bi-LSTM outperformed LSTM in many tasks	Did not test financial-specific data
[4]	2021	Shanghai Stock Exchange	CNN-BiLSTM-Attention	Outperformed vanilla LSTM	More complex model, risk of overfitting
[5]	2023	Financial market data (stock prices; dataset details not extensively specified)	Compared RNN, LSTM, Bi-LSTM, GRU, and Transformer models	GRU and Transformer models demonstrated superior performance in forecasting accuracy	Limited dataset scope; lacks interpretability analysis and broader market validation
[6]	2024	Gold price data (Dec 2017 – Mar 2024, Yahoo Finance)	Compared LSTM and GRU models with recurrent + dense layers	GRU achieved lower error (MSE, MAE) and higher R^2 than LSTM	Focused only on gold; limited models; lacks cross-market validation
[7]	2023	Vietnam stock prices	LSTM vs Bi-LSTM	Bi-LSTM achieved higher accuracy	Dataset limited to a single market
[8]	2019	Survey (various)	Review of ML/DL methods	Comprehensive comparison of forecasting methods	No new empirical results
[9]	2016	Amman Stock Exchange	Regression analysis	Identified key financial factors	No deep learning; traditional approach only
[10]	2022	Financial time series data (various stock market datasets)	ARIMA, GRU, LSTM, Bi-LSTM	Bi-LSTM achieved the highest prediction accuracy	Dataset description limited; evaluation lacks diversity of markets; no interpretability analysis
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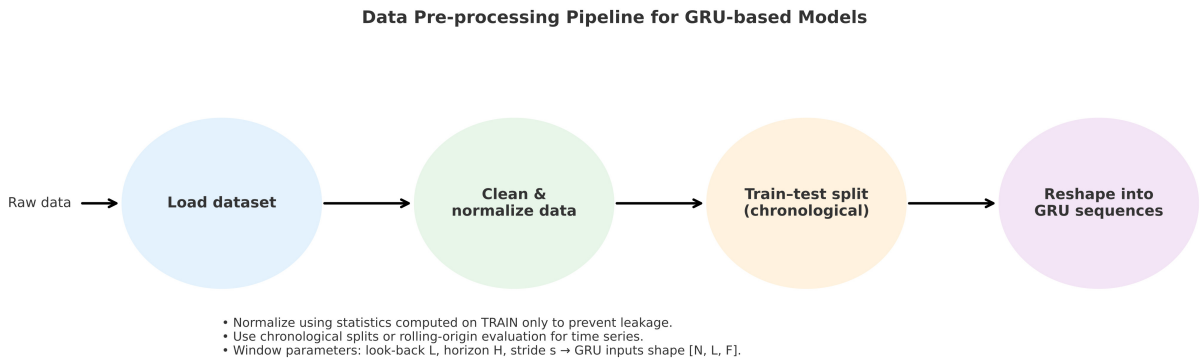


Fig. 1: Data preprocessing overview

for optimization and MSE as the loss function to have consistent and efficient changes in its parameters during training. Through exploiting the capacity to effectively deal with long-term dependencies with low computational overhead, the GRU configuration exhibits high promise for efficient and accurate financial time series prediction.

C. Evaluation Metrics:

In order to determine forecasting performance, error-based and classification-based measures were used. To measure the deviation between the predicted and observed values, Root Mean Squared Error (RMSE) was used as the main performance metric.

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{1}$$

where y_i and $\hat{y}_i$ represent the actual and predicted values, respectively, and N is the total number of samples.

Moreover, directional predictions accuracy was evaluated using precision, recall, and F1-score. The actual and predicted changes of the price were coded into binary data (upward or downward). The following measurements have been calculated

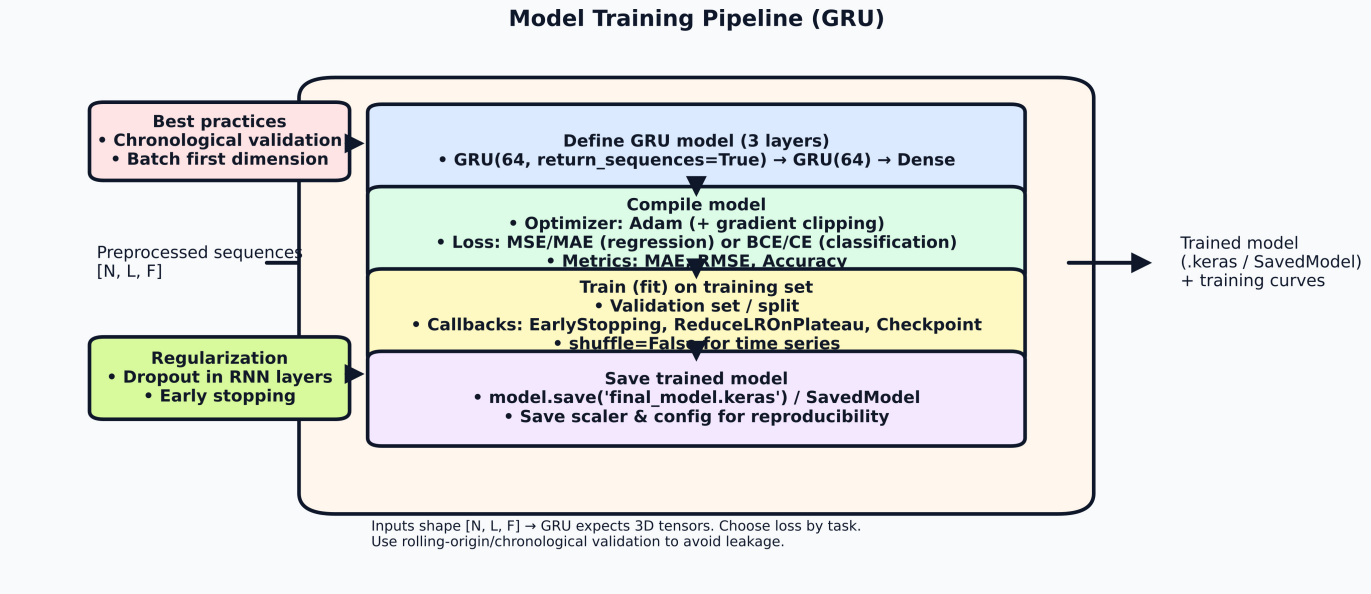


Fig. 2: Model train

using these labels:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

where TP, FP, and FN represent true positives, false positives, and false negatives, respectively. Combined, these measures give a total assessment, giving an overview of the accuracy of both numerical forecasts and the consistency of directional movement predictions.

D. Performance Evaluation

Performance of the models was compared on several fronts such as prediction accuracy, computational efficiency and model complexity. RMSE was used to evaluate numerical error and Precision, recall, and F1 score were applied to assess the model's predictive accuracy. Computational performance was determined in a factor of inference time per sample (ms) and GFLOPS, and model complexity was downed by the number of trainable parameters and model size (MB).

Experimental conditions were kept the same so that all models (LSTM, Bi-LSTM and GRU) are trained in equal conditions. This configuration employed identical data splits, a learning rate of 0.001, batch size, the Adam optimizer, and the Mean Squared Error (MSE) loss function. To have performance measurement that was unbiased, the test dataset was evaluated on the test dataset.

IV. Results and Discussions

A. System Configuration

GRU, LSTM, and BiLSTM models were utilized to forecast financial movements in the BTC and AAPL datasets. For training, we utilized a system equipped with an Intel Core i5 processor, an NVIDIA GTX 1080 Ti GPU, and 16 GB of RAM to ensure computational efficiency. The experimentation was conducted using a Python 3.10 environment alongside TensorFlow 2.x, with help of scikit-learn, pandas, and numpy libraries. Training was carried out in a batch size of 32 and 10 epochs using the Adam optimizer set to a learning rate of 0.001. Each model consisted of 3 stacked RNN layers. Four recurrent layers made up each model, which matched the setup in Tables II and III.

B. Training Parameter Setting

The prediction models, including GRU, LSTM, and BiLSTM, were configured with four recurrent layers and trained on BTC and AAPL datasets to forecast short-term price movements. The optimizer, set to a 0.001 learning rate, was used for all models. It ensures stable and efficient weight updates. Each layer contained a fixed number of hidden units. The trained models were evaluated in terms of inference time, computational complexity (GFLOPS), model size, and classification performance (Precision, Recall, F1-score). As reported in Tables IV and III, the GRU model achieved the smallest parameter count (38,601 for BTC) and fastest inference time (122.362 ms for AAPL), while BiLSTM had the highest parameter count (141,701) and slightly longer inference times. Performance metrics (Tables IV and V) indicate that GRU consistently achieved higher precision and F1 scores compared to LSTM and BiLSTM, highlighting its efficiency in trend classification for both datasets.All training used the

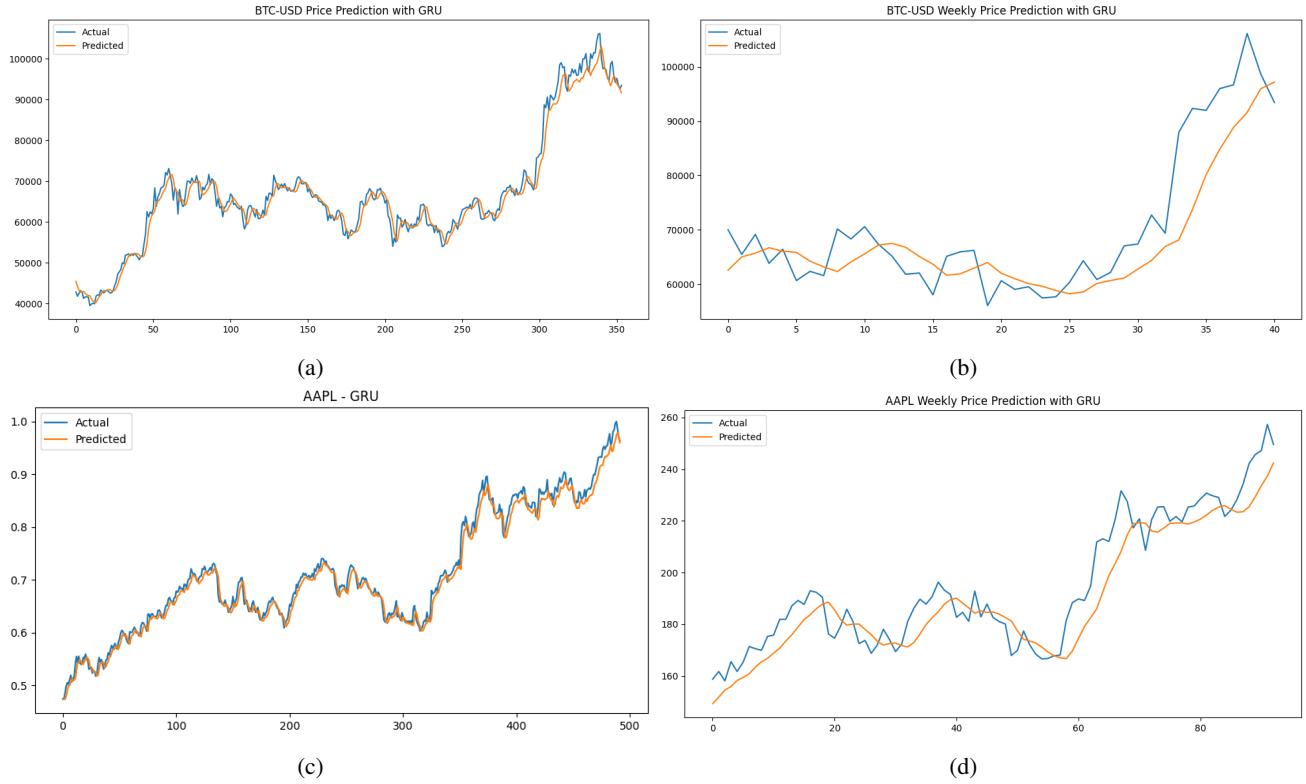


Fig. 3: Four performance metrics curves: (a) precision, (b) recall, (c) F1-score, and (d) RMSE.

chronological train/validation/test split described in Section III to avoid temporal leakage

C. Simulation Results

The results for BTC and AAPL datasets demonstrate the comparative performance of GRU, LSTM, and BiLSTM models in prediction tasks. As shown in Tables IV and III, GRU achieved the fastest inference times (122.362 ms for AAPL and 128.930 ms for BTC) and the smallest model size (0.483 MB for both datasets), where BiLSTM had the highest parameter count (141,701) and slightly longer inference times due to its bidirectional layers. Performance metrics in Tables IV and V reveal that GRU consistently attained the highest Precision (0.99 for BTC, 0.98 for AAPL) and competitive F1-scores (0.59 for BTC, 0.39 for AAPL), while Recall values remained comparable across all models (0.50), indicating similar sensitivity to upward trends. Comparisons with existing results (Table further highlight the superior predictive capability of GRU, with an RMSE of 0.0152, outperforming LSTM (0.0184) and BiLSTM (0.0189). These results suggest that GRU provides an optimal balance between computational efficiency and predictive accuracy for financial trend forecasting. RMSE was utilized for analyzing numerical price prediction (regression). Only directional movement prediction (up = 1, down = 0) based on price differences was assessed using precision, recall, and F1-score. The accuracy of price regression is not determined by these categorization measures.

D. Comparative Discussion

Across all studies, the GRU consistently outperformed LSTM and Bi-LSTM, owing to its simpler gating mechanism, decreased parameter count, and lower computational cost. GRU demonstrated quicker convergence during training and inference periods while maintaining the lowest RMSE values. In contrast, LSTM and Bi-LSTM have more architectural complexity—particularly Bi-LSTM, which analyzes sequences in both forward and backward directions—but this increased complexity did not result in corresponding gains in predicting accuracy. These findings are consistent with previous research in which GRU models outperformed deeper LSTM-based models in scenarios involving noisy or highly volatile financial time series. The experimental findings consequently show that for actual economic forecasting purposes, GRU provides the optimal compromise among precision, reliability, and speed of computation.

V. Conclusion and Future Works

This work gives a benchmark comparison of GRU, LSTM, and Bi-LSTM models rather than offering a novel forecasting technique. The results indicate that the GRU-based approach outperforms the LSTM and Bi-LSTM with regard to predictive accuracy, computational efficiency and general reliability. In particular, GRU demonstrated the best RMSE and maximum accuracy and demonstrated its capability to work with highly volatile and dynamic financial time series. Such results indicate that GRU is a more practical and less heavyweight alterna-

TABLE II: Comparison of deep learning models with 3 RNN layers for BTC

Model	No. of Layers	Parameters	GFLOPS	Optimizer	Learning Rate	Inference Time (ms)	Speed (ms)	Model Size (MB)
BiLSTM	4	141701	0.00002	Adam	0.001	123.646	123.646	1.7
LSTM	4	50851	0.00002	Adam	0.001	125.033	125.033	0.626
GRU	**4**	**38601**	**0.00002**	**Adam**	**0.001**	**128.930**	**123.646**	**0.483**

TABLE III: Comparison of deep learning models with 3 RNN layers for AAPL

Model	No. of Layers	Parameters	GFLOPS	Optimizer	Learning Rate	Inference Time (ms)	Speed (ms)	Model Size (MB)
BiLSTM	4	141701	0.00002	Adam	0.001	122.558	122.558	1.698
LSTM	4	50851	0.00002	Adam	0.001	110.258	110.258	0.625
GRU	**4**	**38601**	**0.00002**	**Adam**	**0.001**	**122.362**	**122.362**	**0.483**

TABLE IV: Comparing GRU, LSTM, and BiLSTM performance for BTC

Model	Precision	Recall	F1-score
BiLSTM	0.98	0.50	0.662
LSTM	0.97	0.50	0.660
GRU	**0.99**	**0.50**	**0.665**

TABLE V: Comparing GRU, LSTM, and BiLSTM performance for Apple stock (AAPL)

Model	Precision	Recall	F1-score
BiLSTM	0.94	0.50	0.653
LSTM	0.90	0.50	0.643
GRU	**0.98**	**0.50**	**0.662**

tive when it comes to the stock and cryptocurrency forecasting tasks. While our current research identifies the dominance of GRU over LSTM and Bi-LSTM in the prediction of Apple stock and Bitcoin prices, a few areas remain where future contributions are needed. Firstly, we plan to integrate methods from Explainable AI, for example SHAP, LIME, and attention visualizations in an attempt to provide more interpretability in our models. It shall facilitate a deeper understanding of the feature contribution of each (e.g., technical metrics, lag returns) towards the above prediction, thereby encouraging more trust and confidence in machine-driven trading platforms. Second, there are other alternative data sources that can be added to the model inputs, including social media indicators, macroeconomic variables, and even news sentiment which may enhance the model's predictive capabilities. Third, future studies may investigate hybrid architectures that integrate GRUs with attention mechanisms, CNNs, or even Transformers, allowing the model to capture immediate patterns as well as extended temporal relationships more effectively. Lastly, we plan to evaluate the proposed approach across diverse markets and extended time horizons to assess its generalizability and robustness.

TABLE VI: The final comparison of GRU, LSTM, and BiLSTM performance with existing results

Model	Precision	RMSE
LSTM	0.90	0.0184
BiLSTM	0.94	0.0189
GRU	**0.99**	**0.0152**

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